Grade 12 Chemistry Final Exam Answer Key

Part A - Multiple Choice

1. b
2. b
3. b
4. b
5. b
6. d
7. c
8. e
9. a
10. d
11. d
12. e
13. c
14. b
15. c
16. b
17. c

Part B - Written Answers (Key Summaries)

1. a) Reactants + Heat -> Products

b) Exothermic

c) Enthalpy diagram shows reactants at higher energy level than products, DeltaH = -257.2 kJ

d) Moles = 23.6 g / 98 g/mol = 0.241 mol; DeltaH = 0.241 mol × -257.2 kJ/mol ~ -62.0 kJ

2. q = mcDeltaT -> m = q / (cDeltaT) = 23400 J / (2.46 × 14.2) ~ 671 g

3. Use q_water = mcDeltaT -> q = 225×4.18×1.7 ~ 1599 J; c = q / (mDeltaT) = 1599 / (55×(99-22)) ~ 0.38 J/g degreesC

4. I2 = 0.450 M -> H2 = 0.450 M; Kc = [H2][I2]/[HI]^2 = 0.2 -> solve for [HI]: [HI] = sqrt((0.45×0.45)/0.2) ~ 0.71

M

5. a) Right shift
b) Left shift
c) No effect
d) Right shift (Le Chatelier)

6. Hesss Law -> Use 3 given reactions to derive: DeltaH = 2×(174.1) 2×(76.6) 2×(285.5) = 27.8 kJ

7. DeltaT = q / (mc) = 4080 / (212×0.385) ~ 50 degreesC

8. a) U: $[Rn]5f^{3}6d^{1}7s^{2}$
b) Bi: $[Xe]4f^{1}5d^{1}6s^{2}6p^{3}$

9. VSEPR shapes, polarity and angles:
FCl_2^+: Bent; NO_3^-: Trigonal planar; PI_5: Trigonal bipyramidal; SO_3: Trigonal planar

10. a) 3p5: Period 3, Group 17, p-block
b) 4d5: Period 5, Group 7, d-block

11. Name: 5-isopropyl-4,6-dimethyloctan-3-ol
Condensed: $C_6H_5CH(CH_3)_2$

12. Drawn line structures for given compounds

13. a) Ketone: Pentan-2-one (Hydration)
b) Alkyl halide addition
c) Ester: Hexyl benzoate
d) Dehydration to alkene

14. Anode: Al, Cathode: Pb
Reaction: $Al + Pb^{2+} \rightarrow Al^{3+} + Pb$
E degreescell = 1.66 (0.13) = 1.79 V

15. a) Acid: Balanced using H_2O and H^+
b) Base: Use OH^- and H_2O for balancing

16. $PbCl_2 \leftrightarrow Pb^{2+} + 2Cl^-$; $K_{sp} = [Pb^{2+}][Cl^-]^2$
Moles Pb = 0.170/207.2 = 8.21e-4

$[Pb^{2+}] = 0.0164$ M; $[Cl^-] = 0.0328$ M

Ksp = 1.76e-5

17. Moles = 5/176 = 0.0284 mol -> [HAsc] = 0.114 M

Use Ka and ICE to find $[HO^+]$ ~ 3e-3 -> pH ~ 2.5

18. a) Strong base -> pH ~ 13.3

b) Use buffer equation: pH = pKa + log([A-]/[HA])

c) Weak acid salt -> pH > 7

d) Half volume = 20 mL, pH = pKa